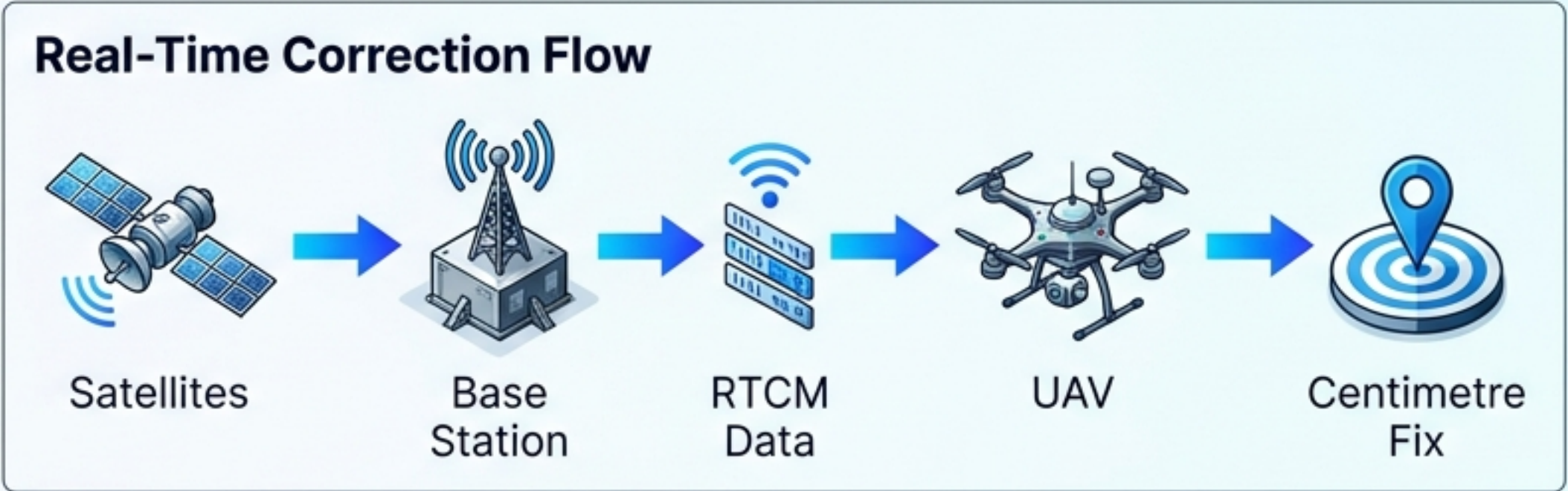


RTK Positioning — The Precision Backbone of UAV Rescue

What It Is & Why It Matters

Problem: Standard GPS provides 1–5m accuracy – entirely insufficient for precision UAV landing in complex disaster zones.

Solution: RTK (Real-Time Kinematic) pairs a fixed Base Station with the UAV Rover to apply real-time RTCM corrections, achieving **2–5cm** accuracy.



Diagnostic Matrix		
GNSS_ONLY	1–5 m accuracy	No correction applied
RTK_FLOAT	0.2–0.5 m accuracy	Correction received, ambiguity unresolved
RTK_FIXED	0.02–0.05 m accuracy	Full RTK fix, ambiguity resolved
CORRECTION_LOST	1–5 m (Degraded)	Correction signal interrupted

BOTTOM LINE: RTK provides the crucial `/uav/rtk_position` topic that Path Planning and QGC Control depend on for successful mission execution.

From Standalone Logic to Full PX4/Gazebo Integration

Two-Phase Simulation Architecture

LEVEL 1: Standalone Architecture

-   4 **ROS 2** nodes, no PX4/Gazebo required.
-   Simulated 50m × 50m square UAV path at 5m/s.
-   Time-based RTK transitions (0–5s → 5–15s → 15s+).
-   Validated core logic, 18-column CSV logs, 5 analysis graphs.

LEVEL 2: PX4/Gazebo Integrated

-   Adds px4_pose_adapter_node.
-   Reads real **Gazebo** navsat via ros_gz_bridge.
-   RTCM correction quality driven (0.0–1.0 score).
-   Simulates signal loss (t=45s) and recovery (t=50s).
-   Real **QGC** mission flown over ETH Zurich **Gazebo** world.
-   22-column CSV logs, 6 graphs (incl. **QGC** cross-validation).

Key Architectural Decision

Utilised **ros_gz_bridge** instead of px4_msgs (not installed) to read **Gazebo** sensor data directly, ensuring a lightweight and robust integration.

What Was Proved – And What Was Not

Results, Validation & Honest Limitations

Positioning Improvement (Level 2)

	Metric	Raw GNSS	RTK Fixed	Improvement
1	Mean Error	2.394 m	0.048 m	
2	Std Dev	1.013 m	0.012 m	
3	Overall	–	–	96.7%

Tested Waypoints (ETH Zurich PX4 World)



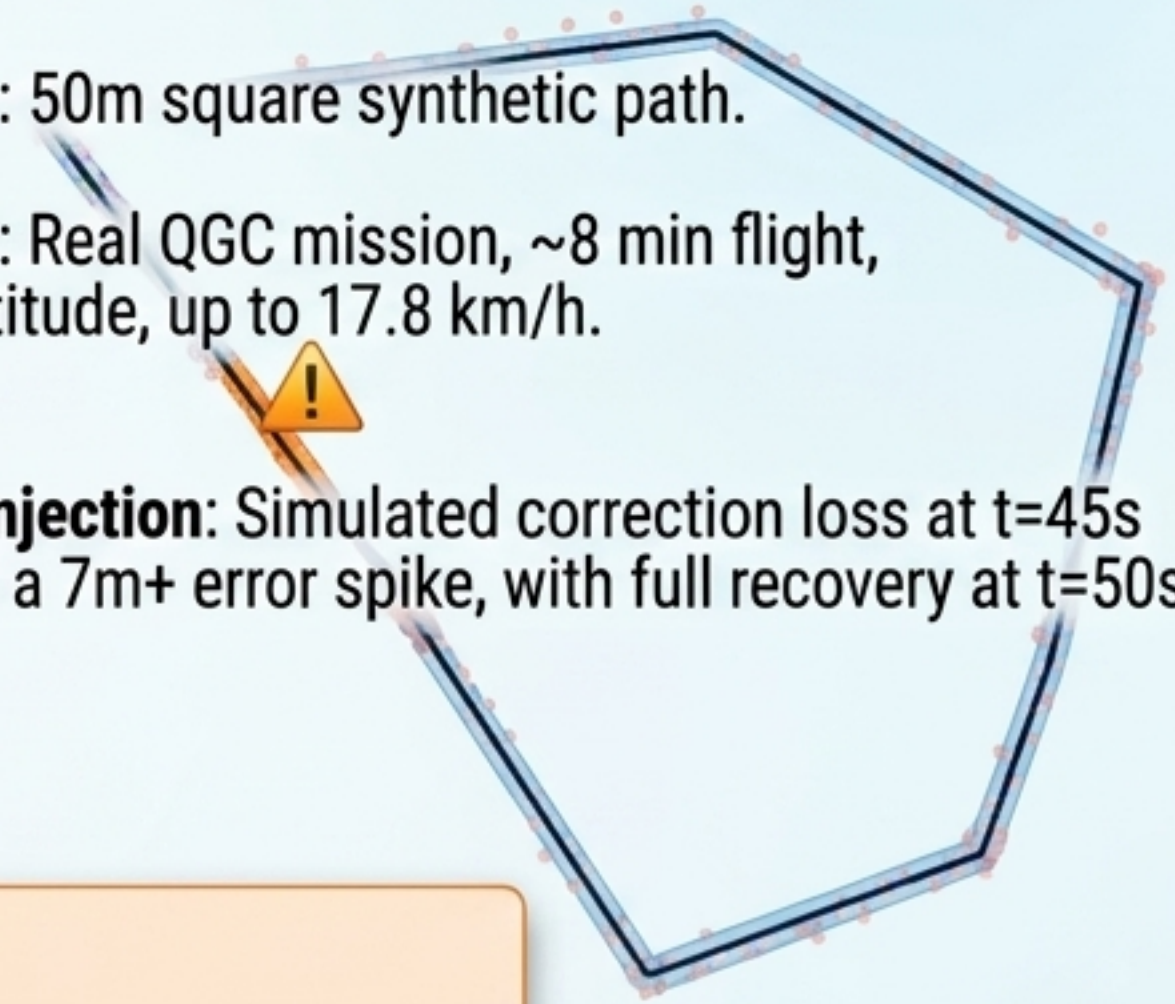
- **Level 1:** 50m square synthetic path.



- **Level 2:** Real QGC mission, ~8 min flight, 50m altitude, up to 17.8 km/h.



- **Event Injection:** Simulated correction loss at t=45s caused a 7m+ error spike, with full recovery at t=50s.



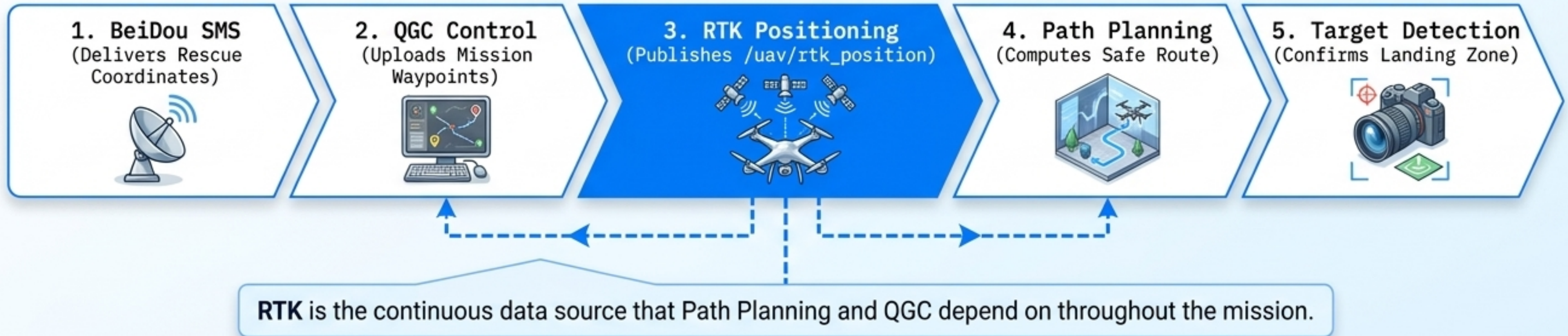
Honest Limitations

- No real RTK hardware; no real RTCM bytes processed.
- Noise is modeled via Gaussian mathematical reduction.
- Correction transitions are time/quality-based, not signal-based.

How RTK Plugs Into the Team System

Integration, Challenges & Next Steps

Full 5-Module Rescue Operation Flow



Challenges Overcome

- WSL2 rendering limits for Gazebo mitigated by using headless simulation.
- Missing `px4_msgs` bypassed via `ros_gz_bridge` workaround.
- Stationary UAV at launch solved by flying a dynamic, real QGC mission.

Next Steps

- Integrate real RTK hardware (BeiDou-compatible u-blox receiver).
- Decode real RTCM binary data.
- Execute full 5-module launch on native Ubuntu 24.04.
- Refine quality thresholds using actual hardware datasheets.